

CHAPTER: 1 - The Skin and Systemic Disease

Professional Clinical Study Guide

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First Edition

Chapter 1

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SUBTITLE: Dermatology — Cutaneous Manifestations of Internal Medicine

SECTION: Recognizing Systemic Disease Through the Skin

The skin is a dynamic mirror of internal pathology. Dermatological findings frequently provide the earliest diagnostic clues to underlying systemic conditions — from autoimmune diseases and infections to malignancies and endocrine disorders. A systematic approach to rash morphology, distribution, evolution, and associated constitutional symptoms is essential for every clinician.

Chapter 2

Clues Suggesting a Rash Is Secondary to Systemic Disease

Clues Suggesting a Rash Is Secondary to Systemic Disease

- Rash associated with **joint pain, fever, weight loss, generalized weakness, shortness of breath, or altered bowel function** — these constitutional symptoms signal systemic involvement and mandate a thorough workup
- Failure to respond to topical therapy — a rash that does not improve after 2–4 weeks of appropriate topical treatment (emollients, topical corticosteroids) should raise suspicion for an underlying systemic etiology
- Erythema secondary to perivascular inflammation — inflammation around dermal blood vessels often indicates a systemic inflammatory or immune-mediated process
- Non-blanching, palpable, purplish, fixed lesions that are painful or blistered — this constellation is classic for **cutaneous vasculitis**. The non-blanching quality indicates extravasation of red blood cells (purpura); palpability reflects vessel wall inflammation; blistering signals severe vascular damage with overlying epidermal necrosis
- Unusual changes in pigmentation or texture — hyperpigmentation (Addison's disease, hemochromatosis), acanthosis nigricans (insulin resistance/malignancy), or velvety thickening can be the earliest visible sign of endocrine, metabolic, or neoplastic disease
- Palpable dermal / subcutaneous nodules — these may result from granulomatous inflammation (sarcoidosis, TB), metastatic deposits (cutaneous metastases from breast, lung, colon, melanoma), lymphoma cutis, or other infiltrative processes. Any new, firm, non-tender dermal nodule warrants biopsy

KEY TEACHING POINTS

- The combination of rash + constitutional symptoms (fever, weight loss, arthralgia, SOB, GI symptoms) should always trigger a systemic workup — do not dismiss these as "just a rash"

- Non-blanching palpable purpura is vasculitis until proven otherwise — the non-blanching nature means RBCs have extravasated; palpable means you can feel it due to vessel wall inflammation
- A rash unresponsive to topical therapy after 2–4 weeks is a red flag for systemic disease — consider biopsy early

MCQ TRAPS & PITFALLS

- "Pyoderma gangrenosum" does NOT imply infection or true gangrene** — it is a neutrophilic dermatosis; cultures are sterile; it does not respond to antibiotics alone
- Erythema nodosum is a septal panniculitis WITHOUT vasculitis** — contrast this with cutaneous vasculitis which involves vessel wall destruction
- Non-blanching purpura is not always vasculitis** — consider embolic phenomena, cholesterol emboli, calciphylaxis in ESRD — but vasculitis is the primary diagnosis to exclude

Chapter 3

SECTION: Characteristic Rashes That Signal Underlying Systemic Disease

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Certain dermatological entities have well-established, high-yield associations with systemic conditions. Recognition of these patterns is essential for both clinical practice and examination purposes. ---

TABLE: Characteristic Rashes and Their Systemic Associations

| Rash / Cutaneous Finding | Systemic Associations | Most Common Cause / Key Notes | |---|---|---| |
****Erythema Multiforme (EM)**** | HSV; Hepatitis B/C; Mycoplasma pneumoniae; drugs | HSV is the #1 infectious trigger; M. pneumoniae especially in children with pneumonia | |
****Pyoderma Gangrenosum**** | Inflammatory Bowel Disease (IBD); Rheumatoid Arthritis (RA); Hepatitis; Hematological malignancy | It is a neutrophilic dermatosis, NOT an infection; IBD (especially UC) is the most common association | |
****Erythema Nodosum**** | IBD; Streptococcal infection; TB; Sarcoidosis; Behçet's disease; drugs | It is the MOST COMMON PANNICULITIS; it is a secondary phenomenon; Streptococcus is the most common identifiable secondary cause | |
****Cutaneous Vasculitis — Purpura**** | Hepatitis B/C; SLE; Lymphoma; Leukemia; Henoch-Schönlein purpura (IgA vasculitis) | Non-blanching palpable purpura; indicates vessel wall inflammation and destruction; Henoch-Schönlein purpura also presents with arthritis, abdominal pain, and renal involvement | ---

QUESTION:

Q: A 34-year-old woman presents with multiple painful, well-demarcated, violaceous-bordered ulcers on her lower legs. She has a history of intermittent bloody diarrhea. Wound cultures are sterile. Which of the following is the most likely diagnosis? A) Necrotizing fasciitis B) Cutaneous tuberculosis C) Pyoderma gangrenosum D) Ecthyma gangrenosum E) Antiphospholipid syndrome cutaneous ulceration ANSWER: C EXPLANATION: Pyoderma gangrenosum is a neutrophilic dermatosis classically associated with inflammatory bowel disease (especially ulcerative colitis). Key diagnostic clues include: (1) rapidly expanding

painful ulcers with violaceous undermined borders, (2) pathergy — lesions may appear at sites of minor trauma, (3) sterile wound cultures, (4) association with systemic diseases including IBD, RA, and hematological malignancies. Necrotizing fasciitis and ecthyma gangrenosum would have positive cultures. Cutaneous TB would show caseating granulomas on biopsy. Antiphospholipid ulcers are less violaceous and more reticulated. SLIDE_REF: ■ Characteristic rashes table ---

Chapter 4

SECTION: Skin Reactions Associated with Infections

SECTION: Skin Reactions Associated with Infections

The skin frequently reacts to infectious agents — both directly (cutaneous invasion) and indirectly (immune-mediated responses). Recognizing the morphology and pattern of infectious skin reactions is critical for diagnosis and management. ---

SUBSECTION: Erythema Multiforme (EM)

Erythema multiforme is an acute, self-limited, immune-mediated hypersensitivity reaction characterized by distinctive target lesions. It represents a spectrum from EM minor (mild, limited mucosal involvement) to EM major (extensive mucosal involvement, systemic symptoms). **Etiology:** • **HSV (Herpes Simplex Virus)** — the #1 infectious trigger, responsible for the majority of recurrent EM cases. EM typically occurs 7–14 days after a herpes outbreak • **Mycoplasma pneumoniae** — especially in children and young adults with atypical pneumonia; can cause severe EM major • **Hepatitis B and C** — less common but well-documented associations • **Drugs** — sulfonamides, anticonvulsants (carbamazepine, phenytoin), NSAIDs, allopurinol **Clinical Features:** • **Target lesions (iris lesions)** — pathognomonic; three concentric zones: central dusky area/pallor, surrounding edematous pale ring, and outer erythematous rim • **Predilection for acral distribution** — dorsal hands, feet, elbows, knees, palms, and soles • Lesions appear in **crops** over 24–48 hours and are typically symmetrical • **EM minor:** limited mucosal involvement (oral mucosa only), no systemic symptoms • **EM major:** extensive mucosal involvement (oral, ocular, genital), fever, malaise **Key Distinction from Stevens-Johnson Syndrome (SJS):** • EM has classic target lesions with well-defined borders; SJS has atypical target lesions or purpuric macules • EM is acral-predominant; SJS is truncal-predominant • SJS involves >2 mucosal surfaces and has significant systemic toxicity • SJS is predominantly drug-induced; EM is predominantly infection-induced (HSV) ---

SUBSECTION: Pyoderma Gangrenosum

Pyoderma gangrenosum is a rare, non-infectious, neutrophilic dermatosis characterized by rapidly progressive, painful ulcers with undermined violaceous borders. Despite its name, it is neither infectious nor gangrenous. **Systemic Associations:** • **Inflammatory Bowel Disease (IBD)** — the most common association, particularly ulcerative colitis (UC); may parallel or precede intestinal disease activity • **Rheumatoid Arthritis (RA)** • **Hepatitis** (especially Hepatitis C) • **Hematological malignancies** — myelodysplastic syndrome (MDS), acute myeloid leukemia (AML), monoclonal gammopathy • **Other:** SLE, sarcoidosis, PAPA syndrome (Pyogenic Arthritis, Pyoderma gangrenosum, Acne) **Clinical Features:** • Begins as a painful pustule, papule, or nodule that rapidly ulcerates • **Undermined violaceous borders** — the hallmark; the edge rolls outward and is deeply erythematous/violaceous • **Pathergy** — new lesions may develop at sites of minor trauma, including surgical wounds, IV sites, and biopsy sites • **Sterile wound cultures** — a critical diagnostic feature distinguishing it from infectious etiologies • Most common location: **lower extremities** (pretibial), but can occur anywhere including stoma sites (peristomal pyoderma gangrenosum) **Management:** • First-line: **Systemic corticosteroids** (prednisolone 0.5–1 mg/kg/day) • Steroid-sparing agents: **ciclosporin**, **mycophenolate mofetil**, **azathioprine** • Biologic therapy: **anti-TNF agents** (infliximab, adalimumab) — particularly effective in IBD-associated cases • Topical:

high-potency corticosteroids, tacrolimus for localized disease • **Avoid surgical debridement** — pathergy may worsen the condition ---

Chapter 5

SUBSECTION: Erythema Nodosum

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Erythema nodosum is the **most common form of panniculitis** — inflammation of the subcutaneous fat. It is a secondary reactive phenomenon rather than a primary disease, and identifying the underlying cause is essential. **Pathology:** • **Septal panniculitis WITHOUT vasculitis** — inflammation is localized to the fibrous septa between fat lobules; this distinguishes it from other panniculitides • **Miescher's microgranulomas** — small collections of histiocytes within the septa; characteristic but not pathognomonic **Systemic Associations (Mnemonic: "SORE SHINS"):** • **S** — Streptococcal infection (most common identifiable secondary cause) • **O** — Oral contraceptive pill / drugs (sulfonamides, penicillins) • **R** — Reactive conditions (IBD, Behçet's disease) • **E** — Eponymous (no specific association) • **S** — Sarcoidosis (especially with bilateral hilar lymphadenopathy — Löfgren syndrome) • **H** — Hansen's disease (leprosy) • **I** — Infections (TB, histoplasmosis, coccidioidomycosis, Yersinia) • **N** — Neoplasms (lymphoma, leukemia) • **S** — Sarcoidosis (repeated for emphasis — very common association) **Clinical Features:** • **Tender, erythematous, subcutaneous nodules** — typically 1–5 cm in diameter • **Pretibial distribution** — bilateral, symmetrical, over the shins; this is the classic and most common location • Lesions are **raised** and **warm** but do NOT ulcerate or scar (distinguishing from other panniculitides) • **Fever, malaise, arthralgia** — constitutional symptoms are common • Lesions evolve through color changes: bright red → purple → yellow-green (resembling a bruise) over 2–6 weeks • **Self-limiting** — resolves spontaneously in 3–6 weeks regardless of treatment **Management:** • Treat the underlying cause • **NSAIDs** (ibuprofen 400mg PO TID, indomethacin 25mg PO TID) for symptomatic relief • **Leg elevation and compression** — reduces edema and pain • **Potassium iodide (KI)** — 300–900 mg/day; effective in refractory cases • Systemic corticosteroids: reserved for severe cases; always exclude TB and fungal infections first (risk of dissemination) ---

KEY TEACHING POINTS

- Erythema nodosum is the MOST COMMON panniculitis and is a REACTIVE phenomenon — always search for an underlying cause
- Pyoderma gangrenosum is a neutrophilic dermatosis, NOT an infection — cultures are sterile; do NOT debride surgically (pathergy risk)
- Erythema multiforme is most commonly triggered by HSV — consider suppressive antiviral therapy (acyclovir 400mg PO BID) for recurrent cases
- The pretibial, bilateral, non-ulcerating, self-resolving nodules of erythema nodosum are distinct from the undermined violaceous ulcers of pyoderma gangrenosum

MCQ TRAPS & PITFALLS

- Erythema nodosum does NOT ulcerate — if there is ulceration, consider other diagnoses (nodular vasculitis, cutaneous TB, pyoderma gangrenosum)
- Pyoderma gangrenosum is NOT pyogenic — the name is a historical misnomer; it is a neutrophilic dermatosis, not an infection
- EM target lesions have 3 zones — central dusky area, pale edematous ring, outer erythematous rim; this distinguishes true targets from the atypical targets of SJS/TEN

- Do not give steroids for erythema nodosum without excluding TB/fungal infection first** — risk of disseminated infection

Chapter 6

IMPORTANT

IMPORTANT

- Always perform a **biopsy** of any atypical or persistent skin lesion — histopathology is often diagnostic
- Pathergy testing** (needle prick test) may support the diagnosis of pyoderma gangrenosum or Behçet's disease, but is neither sensitive nor specific
- Henoch-Schönlein purpura (IgA vasculitis)** is the most common vasculitis in children; presents with the tetrad of palpable purpura (lower extremities/buttocks), arthritis/arthralgia, abdominal pain, and renal involvement (IgA nephropathy)

OSCE PEARLS

- When examining a patient with purpura, always **demonstrate blanching** — press a glass slide firmly against the lesion; non-blanching indicates extravasated RBCs (purpura/vasculitis)
- For erythema nodosum, examine the **pretibial areas bilaterally** and ask about recent sore throat (streptococcal infection), chest symptoms (sarcoidosis), and bowel symptoms (IBD)
- For pyoderma gangrenosum, describe the **undermined violaceous border** and check for **pathergy** at IV sites or surgical scars
- Always ask about **medication history** — many cutaneous drug reactions mimic systemic disease rashes

QUESTION:

Q: A 28-year-old man presents with a 5-day history of bilateral, tender, erythematous nodules over his shins. He reports a sore throat 2 weeks ago. Chest X-ray shows bilateral hilar lymphadenopathy. ESR is elevated at 68 mm/hr. Which of the following is the most likely unifying diagnosis? A) Streptococcal erythema nodosum B) Löfgren syndrome (acute sarcoidosis) C) Behçet's disease D) Inflammatory bowel disease E) Tuberculosis ANSWER: B EXPLANATION: Löfgren syndrome is the acute presentation of sarcoidosis characterized by the triad of: (1) bilateral hilar lymphadenopathy (BHL), (2) erythema nodosum, and (3) fever/arthralgia. This patient has all three features plus elevated ESR. While streptococcal infection is the most common cause of erythema nodosum overall, the presence of BHL makes sarcoidosis the unifying diagnosis. Löfgren syndrome carries a good prognosis with spontaneous remission in >80% of cases. Behçet's disease would present with oral/genital ulcers and uveitis. IBD would have GI symptoms. TB would show unilateral hilar lymphadenopathy (more common) and positive Mantoux/IGRA. SLIDE_REF: ■ Erythema nodosum section ---

Chapter 7

QUESTION:

QUESTION:

Q: A 45-year-old man with known ulcerative colitis presents with a rapidly enlarging, painful ulcer on his left shin. The ulcer has a violaceous, undermined border and a necrotic base. Wound swabs show no growth. He recently had a skin biopsy of his forearm that developed a similar lesion at the biopsy site. What is the most appropriate next step in management? A) Surgical debridement of the ulcer B) Intravenous antibiotics targeting *Pseudomonas* C) Oral prednisolone 1 mg/kg/day D) Topical antifungal cream E) Incisional biopsy with tissue culture ANSWER: C EXPLANATION: This presentation is classic for pyoderma gangrenosum: (1) patient with UC (most common systemic association), (2) rapidly progressive painful ulcer with undermined violaceous border, (3) sterile wound cultures, (4) pathergy (new lesion at biopsy site). First-line treatment is systemic corticosteroids (oral prednisolone 0.5–1 mg/kg/day). Surgical debridement (A) is CONTRAINDICATED due to pathergy — it would worsen the condition. IV antibiotics (B) are not indicated as cultures are sterile. Topical antifungals (D) are irrelevant. While biopsy (E) may be performed to exclude other diagnoses, it should be done cautiously given pathergy risk, and treatment should not be delayed. SLIDE_REF: ■ Pyoderma gangrenosum section ---

QUESTION:

Q: A 19-year-old woman presents with recurrent episodes of target lesions on her hands and feet, each episode occurring 1–2 weeks after a cold sore. The lesions have three concentric zones of color change. She is otherwise well. What is the most appropriate long-term management? A) Oral acyclovir 400mg twice daily B) Oral prednisolone 40mg daily C) Topical betamethasone valerate 0.1% D) Oral dapsone 100mg daily E) Intravenous immunoglobulin ANSWER: A EXPLANATION: This patient has recurrent erythema multiforme (EM) triggered by HSV (herpes labialis/cold sores). The three-zone target lesions on acral sites are pathognomonic. Since HSV is the trigger, long-term suppressive antiviral therapy with acyclovir 400mg PO BID is the most appropriate management to prevent recurrent episodes. This is supported by clinical evidence showing significant reduction in EM recurrence with chronic acyclovir prophylaxis. Systemic steroids (B) are not recommended for EM. Topical steroids (C) have limited benefit in established lesions. Dapsone (D) is used in some chronic EM cases but is not first-line. IVIG (E) is reserved for severe SJS/TEN, not EM. SLIDE_REF: ■ Erythema multiforme section ---

SECTION: Cutaneous Vasculitis

Cutaneous vasculitis refers to inflammation and destruction of blood vessel walls within the skin. It may be limited to the skin (cutaneous small vessel vasculitis) or represent a manifestation of systemic vasculitis with multi-organ involvement. ---

Chapter 8

SUBSECTION: Classification and Pathophysiology

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****Classification by Vessel Size:**** | Vessel Size | Examples | Key Features | ---|---|---| ****Small vessel**** | Henoch-Schönlein purpura (IgA vasculitis), cutaneous leukocytoclastic vasculitis, urticarial vasculitis | Palpable purpura; lower extremities; leukocytoclasia (nuclear debris from neutrophils) | ****Medium vessel**** | Polyarteritis nodosa (PAN), Kawasaki disease | Nodular lesions, livedo reticularis, subcutaneous aneurysms; can cause digital gangrene | ****Large vessel**** | Giant cell arteritis, Takayasu arteritis | Absent pulses, claudication, visual symptoms; skin findings less prominent | ****Pathophysiology:**** • ****Immune complex deposition**** — the most common mechanism; antigen-antibody complexes deposit in vessel walls, activate complement (C3a, C5a), recruit neutrophils, and cause vessel wall destruction • ****ANCA-associated**

vasculitis** — anti-neutrophil cytoplasmic antibodies activate neutrophils, causing them to degranulate and damage vessel walls; includes granulomatosis with polyangiitis (GPA), microscopic polyangiitis (MPA), eosinophilic granulomatosis with polyangiitis (EGPA) ---

SUBSECTION: Clinical Features of Cutaneous Small Vessel Vasculitis

- Palpable purpura** — the hallmark; non-blanching, raised, purple lesions predominantly on the lower extremities and dependent areas (due to hydrostatic pressure)
- Distribution**: gravity-dependent areas — legs, ankles, feet; in ambulatory patients, the lower legs; in bedridden patients, the sacral area
- Lesion evolution**: individual lesions typically resolve over 2–4 weeks but new lesions may continue to appear in crops
- Associated symptoms**: burning or pruritus (more common than pain); in systemic vasculitis — fever, arthralgia, myalgia, abdominal pain, renal involvement (hematuria/proteinuria)
- Severe forms**: hemorrhagic bullae, ulceration, necrosis — indicate severe vessel wall destruction

SUBSECTION: Henoch-Schönlein Purpura (IgA Vasculitis)

Henoch-Schönlein purpura (HSP) is the **most common systemic vasculitis in children** (peak age 4–6 years) but can occur in adults. It is an IgA-mediated small vessel vasculitis. **Classic Tetrad:** 1. **Palpable purpura** — non-blanching, predominantly on lower extremities and buttocks 2. **Arthritis/arthralgia** — non-deforming, affecting knees and ankles; oligoarticular 3. **Abdominal pain** — colicky; may be associated with GI bleeding (intussusception in children) 4. **Renal involvement** — IgA nephropathy; ranges from microscopic hematuria to nephritic/nephrotic syndrome **Triggers:** • Upper respiratory tract infections (especially Group A Streptococcus) • Drugs, foods, insect bites (less common) **Investigations:** • **Skin biopsy**: leukocytoclastic vasculitis with **IgA deposition** on direct immunofluorescence (DIF) — this is diagnostic • **Urinalysis**: hematuria, proteinuria — monitor for renal involvement • **Serum IgA**: elevated in ~50% of cases • **ANCA**: negative (distinguishes from ANCA-associated vasculitis) **Management:** • **Supportive** — most cases in children are self-limiting • **NSAIDs** for arthralgia (if no renal involvement) • **Systemic corticosteroids** (prednisolone 1 mg/kg/day) — for severe GI symptoms or significant renal involvement • **Renal monitoring** — urinalysis for 6 months after diagnosis (delayed renal involvement can occur) • **Immunosuppressives** (cyclophosphamide, azathioprine) — for progressive renal disease ---

Chapter 9

KEY TEACHING POINTS

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- Cutaneous vasculitis presents as non-blanching palpable purpura in gravity-dependent areas — the palpability reflects vessel wall inflammation
- HSP (IgA vasculitis) is the most common vasculitis in children — the classic tetrad is purpura, arthritis, abdominal pain, and renal involvement
- Skin biopsy with direct immunofluorescence is essential — IgA deposition confirms HSP; pauci-immune pattern suggests ANCA-associated vasculitis
- Always check urinalysis in any patient with cutaneous vasculitis — renal involvement significantly alters management and prognosis

MCQ TRAPS & PITFALLS

- HSP renal involvement may be delayed** — up to 6 months after the initial presentation; monitor urinalysis for at least 6 months
- Palpable purpura in an adult is more likely to be systemic vasculitis** — unlike children where HSP is common, adults with palpable purpura need thorough investigation for underlying systemic vasculitis, malignancy, or infection
- Leukocytoclastic vasculitis is a histological pattern, not a diagnosis** — it describes the finding of neutrophil nuclear debris in vessel walls; the underlying cause must be identified

IMPORTANT

- ANCA testing** (p-ANCA/MPO and c-ANCA/PR3) should be performed in all adults with cutaneous vasculitis to exclude ANCA-associated vasculitis (GPA, MPA, EGPA)
- Cryoglobulinemia** should be considered in patients with palpable purpura, especially with Hepatitis C association — test cryoglobulins and complement levels (C4 is characteristically low)
- Cholesterol emboli syndrome** can mimic cutaneous vasculitis — look for livedo reticularis, blue toes, eosinophilia, and elevated creatinine; often follows vascular procedures or anticoagulation

Chapter 10

TABLE: Comparison of Vasculitis Types

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Feature	HSP (IgA Vasculitis)	Cutaneous Leukocytoclastic Vasculitis	ANCA-Associated Vasculitis	Polyarteritis Nodosa
Vessel size	Small	Small	Small-medium	Medium
Age group	Children (4–6 years)	Any age	Adults >50 years	Adults 40–60 years
Key serology	Elevated serum IgA	None specific	c-ANCA (GPA) or p-ANCA (MPA)	Hepatitis B association (30%)
Skin findings	Palpable purpura (legs/buttocks)	Palpable purpura (dependent areas)	Purpura, ulcers, nodules	Livedo reticularis, subcutaneous nodules, digital gangrene
Renal involvement	IgA nephropathy	Rare	Pauci-immune GN (rapidly progressive)	Renal artery aneurysms, renovascular hypertension
GI involvement	Abdominal pain, intussusception	Rare	Rare	Mesenteric ischemia, bowel infarction
Diagnostic test	Skin biopsy: IgA on DIF	Skin biopsy: leukocytoclasia	ANCA serology + biopsy	Angiography: microaneurysms

QUESTION:

Q: A 6-year-old boy presents with a 3-day history of a non-blanching, purpuric rash on his legs and buttocks. He also has painful, swollen knees and intermittent colicky abdominal pain. Urinalysis shows 2+ blood and 1+ protein. His mother reports he had a sore throat 2 weeks ago. Which of the following is the most likely diagnosis? A) Kawasaki disease B) Henoch-Schönlein purpura C) Juvenile idiopathic arthritis D) Acute post-streptococcal glomerulonephritis E) Meningococemia ANSWER: B EXPLANATION: This child presents with the classic tetrad of Henoch-Schönlein purpura (HSP): (1) palpable purpura on lower extremities and buttocks, (2) arthritis (swollen knees), (3) abdominal pain, and (4) renal involvement (hematuria and proteinuria). The preceding sore throat is a common trigger (Group A Streptococcus). Kawasaki disease (A) presents with fever >5 days, conjunctival injection, strawberry tongue, and desquamation — not purpura. JIA (C) does not cause purpura or renal involvement. Post-streptococcal GN (D) causes nephritic syndrome but

not purpura or arthritis. Meningococcemia (E) causes rapidly progressive purpura fulminans with septic shock, not a self-limited course with arthritis and abdominal pain. SLIDE_REF: ■ Henoch-Schönlein purpura section ---

QUESTION:

Q: A 55-year-old woman presents with palpable purpura on her lower legs, arthralgia, and microscopic hematuria. c-ANCA (PR3) is positive. Chest X-ray shows pulmonary nodules. Which of the following is the most likely diagnosis? A) Henoch-Schönlein purpura B) Granulomatosis with polyangiitis (GPA) C) Microscopic polyangiitis (MPA) D) Eosinophilic granulomatosis with polyangiitis (EGPA) E) Polyarteritis nodosa ANSWER: B EXPLANATION: Granulomatosis with polyangiitis (GPA, formerly Wegener's) is characterized by the triad of: (1) upper respiratory tract involvement (sinusitis, epistaxis, saddle nose deformity), (2) lower respiratory tract involvement (pulmonary nodules, cavitating lesions), and (3) glomerulonephritis. c-ANCA (PR3) is positive in >90% of generalized GPA cases. The combination of cutaneous vasculitis (palpable purpura), pulmonary nodules, renal involvement, and c-ANCA positivity is diagnostic. MPA (C) is associated with p-ANCA (MPO), not c-ANCA. EGPA (D) is associated with asthma, eosinophilia, and p-ANCA. HSP (A) occurs in children and is IgA-mediated. PAN (E) does not cause pulmonary involvement or ANCA positivity. SLIDE_REF: ■ Comparison of vasculitis types table ---

Chapter 11

SECTION: Other Important Cutaneous Manifestations of Systemic Disease

SECTION: Other Important Cutaneous Manifestations of Systemic Disease

SUBSECTION: Acanthosis Nigricans

Acanthosis nigricans presents as **velvety, hyperpigmented, thickened skin** in intertriginous areas (axillae, groin, neck, inframammary folds). It is a cutaneous marker of insulin resistance and, in rare cases, internal malignancy. **Benign Associations:** • **Insulin resistance / Type 2 Diabetes Mellitus** — the most common association; hyperinsulinemia stimulates keratinocyte and fibroblast growth factor receptors • **Obesity** — very common; often the sole association • **Polycystic ovary syndrome (PCOS)** • **Endocrine disorders:** Cushing's syndrome, acromegaly, hypothyroidism • **Drugs:** nicotinic acid, oral contraceptive pill, corticosteroids **Malignant Acanthosis Nigricans:** • Associated with **gastric adenocarcinoma** (most common malignancy), also other GI malignancies, lymphoma • **Key distinguishing features:** rapid onset, extensive involvement (including mucous membranes — lips, oral mucosa, palms), associated with weight loss and other paraneoplastic signs • **Leser-Trélat sign:** sudden eruption of multiple seborrheic keratoses — another paraneoplastic marker associated with GI malignancies ---

SUBSECTION: Dermatomyositis

Dermatomyositis (DM) is an idiopathic inflammatory myopathy with characteristic skin findings that often precede muscle weakness by months. It is associated with malignancy in approximately 15–25% of adult cases. **Characteristic Skin Findings:** • **Heliotrope rash** — violaceous/lilac discoloration of the upper eyelids with periorbital edema; pathognomonic • **Gotttron's papules** — erythematous/violaceous papules over the MCP and IP joints; pathognomonic • **Gotttron's sign** — erythematous/violaceous patches over

knuckles, elbows, knees, and medial malleoli • **Shawl sign** — erythema over the posterior neck and shoulders (shawl distribution) • **V-sign** — erythema over the anterior neck and chest in a V-shape • **Mechanic's hands** — hyperkeratotic, fissured skin on the lateral fingers and palms (more common in anti-synthetase syndrome) • **Nail fold changes** — periungual telangiectasia, ragged cuticles • **Malignancy Screening**: • All adults with newly diagnosed DM should undergo **age-appropriate malignancy screening** • Most commonly associated malignancies: **ovarian, lung, gastric, colorectal, lymphoma** • Risk is highest in the **first 3 years** after diagnosis • **Anti-TIF1- γ** (transcription intermediary factor 1-gamma) antibodies and **anti-NXP2 antibodies** are associated with increased malignancy risk in DM • **Investigations**: • **CK** (creatinine kinase): elevated (may be normal in amyopathic DM) • **EMG**: myopathic changes • **MRI**: muscle edema (inflammation) • **Muscle biopsy**: perifascicular atrophy, perivascular inflammation • **Skin biopsy**: interface dermatitis similar to lupus • **ANA**: positive in ~60–80% • **Myositis-specific antibodies**: anti-Mi-2, anti-MDA5, anti-TIF1- γ , anti-NXP2, anti-Jo-1 (anti-synthetase) • **Management**: • **Systemic corticosteroids** (prednisolone 1 mg/kg/day) — first-line • **Steroid-sparing agents**: methotrexate, azathioprine, mycophenolate mofetil • **IVIg** — for refractory disease or severe skin involvement • **Rituximab** — for refractory cases • **Hydroxychloroquine** — for skin-predominant disease • **Physiotherapy** — essential to maintain muscle function • **Sun protection** — photosensitivity is common ---

Chapter 12

SUBSECTION: Systemic Lupus Erythematosus (SLE) — Cutaneous Manifestations

SUBSECTION: Systemic Lupus Erythematosus (SLE) — Cutaneous Manifestations

SLE is a multisystem autoimmune disease with diverse cutaneous manifestations. Skin involvement occurs in >80% of SLE patients and may be the presenting feature. • **Acute Cutaneous Lupus**: • **Malar (butterfly) rash** — erythematous rash over the cheeks and nasal bridge, sparing the nasolabial folds; photosensitive; may be flat or raised • **Generalized maculopapular rash** — widespread erythematous rash, often photosensitive • **Bullous SLE** — rare; tense blisters on erythematous base • **Subacute Cutaneous Lupus (SCLE)**: • **Annular or psoriasiform lesions** — non-scarring, photosensitive, on sun-exposed areas (upper trunk, arms, neck) • Associated with **anti-Ro/SSA antibodies** (70–90%) • Less likely to have severe renal involvement than other forms • Drug-induced SCLE: hydrochlorothiazide, terbinafine, calcium channel blockers, NSAIDs • **Chronic Cutaneous Lupus (Discoid Lupus)**: • **Discoid plaques** — well-defined, erythematous, scaly plaques with central atrophy, follicular plugging, and scarring alopecia when on the scalp • **Scarring** — a key feature distinguishing discoid lupus from SCLE • 5–10% of isolated discoid lupus patients may progress to SLE • **Lupus profundus (lupus panniculitis)** — deep, firm subcutaneous nodules that may overlie as discoid lupus; can cause significant cosmetic deformity • **Other Cutaneous Features of SLE**: • **Livedo reticularis** — associated with antiphospholipid antibodies • **Raynaud's phenomenon** — often the earliest symptom • **Oral/nasal ulcers** — painless (distinguishing from Behçet's which are painful) • **Non-scarring alopecia** — diffuse thinning ("lupus hair") • **Photosensitivity** — a formal ACR/EULAR classification criterion ---

KEY TEACHING POINTS

- Acanthosis nigricans in a non-obese patient with rapid onset should raise suspicion for underlying malignancy (especially gastric adenocarcinoma)
- Dermatomyositis skin findings (heliotrope rash, Gottron's papules) may precede muscle weakness by months — screen for malignancy in all adult patients

- Malar rash in SLE spares the nasolabial folds — this distinguishes it from other facial rashes (e.g., seborrheic dermatitis, rosacea)
- Discoid lupus causes scarring; subacute cutaneous lupus does NOT scar — this distinction has prognostic significance

MCQ TRAPS & PITFALLS

- Amyopathic dermatomyositis** (skin findings without muscle weakness) still carries a malignancy risk — do not skip cancer screening
- Anti-dsDNA antibodies** are specific for SLE and correlate with renal disease (lupus nephritis); **anti-Smith (Sm)** antibodies are the most specific for SLE but less sensitive
- SCLE is strongly associated with anti-Ro/SSA antibodies** — these can cross the placenta and cause neonatal lupus with congenital heart block
- Discoid lupus scarring alopecia is permanent** — early treatment is essential to prevent irreversible hair loss

Chapter 13

IMPORTANT

IMPORTANT

- Calcinosis cutis** is more characteristic of dermatomyositis (especially juvenile DM) than SLE — calcium deposits in subcutaneous tissue
- Livedo reticularis + recurrent thrombosis = antiphospholipid syndrome** — check lupus anticoagulant, anticardiolipin antibodies, anti-β2 glycoprotein I
- Rowell's syndrome**: erythema multiforme-like lesions in SLE patients — a recognized but rare association

OSCE PEARLS

- When examining for malar rash, **ask the patient to look up** — this flattens the nasolabial fold and makes the sparing more apparent
- For dermatomyositis, examine the **eyelids (heliotrope), knuckles (Gottron's papules), elbows, knees, and nail folds** systematically
- In suspected SLE, always check **urinalysis** (lupus nephritis) and **blood counts** (cytopenias are common and part of classification criteria)
- Photosensitivity history** is crucial — ask if the rash worsens with sun exposure

QUESTION:

Q: A 52-year-old woman presents with a 3-month history of progressive violaceous discoloration of her upper eyelids and erythematous papules over her knuckles. She also reports difficulty climbing stairs and rising from a chair. CK is elevated at 2,500 U/L. Chest CT shows a 3-cm right lower lobe mass. Which of the following is the most likely unifying diagnosis? A) Systemic lupus erythematosus B) Dermatomyositis with associated malignancy C) Polymyalgia rheumatica D) Inclusion body myositis E) Drug-induced myopathy
ANSWER: B EXPLANATION: This patient has dermatomyositis (DM) with: (1) heliotrope rash (violaceous

eyelid discoloration), (2) Gottron's papules (erythematous papules over knuckles), (3) proximal muscle weakness (difficulty climbing stairs/rising from chair), (4) elevated CK (2,500 U/L). The presence of a lung mass on CT raises concern for paraneoplastic DM — malignancy is found in 15–25% of adult DM cases, most commonly lung, ovarian, gastric, and colorectal. SLE (A) would not typically cause this combination of skin findings and elevated CK. Polymyalgia rheumatica (C) causes proximal pain/stiffness but not skin rash or elevated CK. Inclusion body myositis (D) is not associated with characteristic skin findings. Drug-induced myopathy (E) would not cause the skin findings described. SLIDE_REF: ■ Dermatomyositis section ---

Chapter 14

QUESTION:

QUESTION:

Q: A 30-year-old woman presents with a photosensitive, non-scalery, annular rash on her upper back and arms. She has no scarring or atrophy. Anti-Ro/SSA antibodies are positive. Urinalysis is normal. Which of the following is the most likely diagnosis? A) Systemic lupus erythematosus with lupus nephritis B) Subacute cutaneous lupus erythematosus C) Discoid lupus erythematosus D) Dermatomyositis E) Psoriasis ANSWER: B EXPLANATION: Subacute cutaneous lupus erythematosus (SCLE) is characterized by: (1) non-scarring, photosensitive, annular or psoriasiform lesions on sun-exposed areas, (2) strong association with anti-Ro/SSA antibodies (70–90%), (3) less severe systemic disease (normal urinalysis makes significant renal involvement unlikely). The absence of scarring distinguishes SCLE from discoid lupus (C). While SCLE can occur within SLE, the absence of renal involvement and the specific presentation favor isolated SCLE. Dermatomyositis (D) would have heliotrope rash and Gottron's papules. Psoriasis (E) would have well-demarcated, scaly plaques and is not typically annular or photosensitive. SLIDE_REF: ■ SLE cutaneous manifestations section ---

SECTION: Summary and High-Yield Associations

TABLE: High-Yield Cutaneous-Systemic Disease Associations

| Cutaneous Finding | Think of... | Key Investigation | |---|---|---| | Non-blanching palpable purpura | Cutaneous vasculitis / HSP | Skin biopsy + DIF, urinalysis, ANCA | | Violaceous undermined ulcer | Pyoderma gangrenosum | Biopsy (exclude infection), wound culture | | Pretibial tender nodules | Erythema nodum | CXR, ASO titre, Mantoux, ACE level | | Target lesions (acral) | Erythema multiforme | HSV serology, clinical diagnosis | | Velvety hyperpigmentation (axillae) | Acanthosis nigricans | Glucose/HbA1c; if rapid onset → malignancy screen | | Heliotrope rash + Gottron's papules | Dermatomyositis | CK, EMG, MRI, myositis antibodies, malignancy screen | | Malar rash (sparing nasolabial folds) | SLE | ANA, anti-dsDNA, anti-Smith, urinalysis, complement | | Livedo reticularis | Antiphospholipid syndrome / vasculitis | Lupus anticoagulant, anticardiolipin, anti-β2 GP1 | | Annular non-scarring photosensitive rash | SCLE | Anti-Ro/SSA, anti-La/SSB |

Chapter 15

KEY TEACHING POINTS

KEY TEACHING POINTS

- The skin is the most accessible organ for diagnosing systemic disease — always examine the skin thoroughly in any unwell patient
- Non-blanching palpable purpura = vasculitis until proven otherwise** — investigate with biopsy, DIF, ANCA, and urinalysis
- Pyoderma gangrenosum is a clinical diagnosis** supported by sterile cultures, pathergy, and association with systemic disease — do not debride
- Erythema nodosum is reactive** — always search for underlying cause (streptococcal infection, sarcoidosis, IBD, TB, drugs)
- Dermatomyositis in adults mandates malignancy screening** — the skin may be the first clue to an underlying cancer
- SLE skin findings are photosensitive** — sun exposure is a major trigger; advise sun protection in all patients

IMPORTANT

IMPORTANT

- ****ICD-10 codes for reference****: L51.0 (Erythema multiforme), L88 (Pyoderma gangrenosum), L52 (Erythema nodosum), L93.2 (Dermatomyositis), M32.9 (SLE), L90.8 (Discoid lupus)
- ****Biopsy is the single most important investigation**** in undiagnosed skin lesions — always consider it early
- ****Direct immunofluorescence (DIF)**** is essential for diagnosing immune-mediated skin diseases (HSP, SLE, vasculitis)
- ****Drug history is critical**** — many cutaneous manifestations of systemic disease can also be drug-induced

QUESTION:

Q: A 60-year-old man presents with a 2-week history of a non-blanching, purpuric rash on his lower legs. He also reports fatigue, weight loss, and intermittent fevers. Full blood count shows anemia and thrombocytopenia. ANCA is negative. Serum protein electrophoresis shows an IgG kappa paraprotein. What is the most likely underlying diagnosis? A) Henoch-Schönlein purpura B) Cryoglobulinemic vasculitis C) Monoclonal gammopathy-associated vasculitis D) Granulomatosis with polyangiitis E) Cutaneous small vessel vasculitis (idiopathic) ANSWER: C EXPLANATION: This patient presents with cutaneous vasculitis (non-blanching palpable purpura) in the context of constitutional symptoms (fatigue, weight loss, fevers), cytopenias (anemia, thrombocytopenia), and a monoclonal paraprotein (IgG kappa on SPEP). This constellation suggests an underlying hematological malignancy or monoclonal gammopathy-associated vasculitis. Monoclonal gammopathy can cause vasculitis through cryoglobulin deposition, immune complex formation, or direct vascular damage. HSP (A) occurs in children. Cryoglobulinemic vasculitis (B) is possible but cryoglobulins are not mentioned. GPA (D) would typically be ANCA-positive. Idiopathic cutaneous vasculitis (E) would not explain the paraprotein or constitutional symptoms. Further investigation should include bone marrow biopsy to evaluate for myeloma or lymphoproliferative disorder. SLIDE_REF: ■ Cutaneous vasculitis section ---

QUESTION:

QUESTION:

Q: A 25-year-old woman presents with a photosensitive rash over her cheeks and nose that spares the nasolabial folds. She also has painless oral ulcers, arthralgia, and fatigue. Lab results show: ANA positive (1:640, homogeneous pattern), anti-dsDNA positive, C3 and C4 low, and 2+ proteinuria. Which of the following is the most likely diagnosis? A) Subacute cutaneous lupus erythematosus B) Systemic lupus erythematosus C) Dermatomyositis D) Mixed connective tissue disease E) Drug-induced lupus ANSWER: B EXPLANATION: This patient meets multiple criteria for systemic lupus erythematosus (SLE) per the 2019 ACR/EULAR classification criteria: (1) malar rash (photosensitive, sparing nasolabial folds), (2) oral ulcers (painless), (3) arthritis (arthralgia), (4) renal involvement (proteinuria), (5) positive ANA, (6) positive anti-dsDNA (highly specific), (7) low complement (C3, C4 — indicates active disease and consumption). The combination of anti-dsDNA positivity, low complement, and proteinuria strongly suggests lupus nephritis. SCLE (A) would have anti-Ro/SSA antibodies and typically no renal involvement. Dermatomyositis (C) would have heliotrope rash and Gottron's papules. MCTD (D) is associated with anti-U1 RNP antibodies. Drug-induced lupus (E) is associated with anti-histone antibodies and typically does not cause renal disease or anti-dsDNA positivity. SLIDE_REF: ■ SLE cutaneous manifestations section ---

CHAPTER: 1 - The Skin and Systemic Disease

SUBTITLE: End of Chapter Summary

This chapter has covered the essential dermatological manifestations of systemic disease. The key principle throughout is that ****the skin is a window to internal pathology**** — careful examination and recognition of characteristic patterns can lead to early diagnosis and management of serious underlying conditions.

****Remember:****

- Always consider systemic disease when a rash is associated with constitutional symptoms
- Non-blanching palpable purpura = vasculitis — investigate thoroughly
- Pyoderma gangrenosum is NOT an infection — do not debride
- Erythema nodosum is reactive — find the cause
- Dermatomyositis in adults = malignancy screen
- SLE skin findings are photosensitive — protect from sun
- Biopsy early when the diagnosis is uncertain